

Dept. for Speech, Music and Hearing
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**Experiments on vocal tract
transfer**

Fujimura, O. and Lindqvist-Gauffin, J.

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I. SPEECH PRODUCTION

A. EXPERIMENTS ON VOCAL TRACT TRANSFER

The sweep-tone study of the vocal tract response, which was reported on in previous reports⁽⁴⁾⁽⁵⁾, is being continued. Some of the further results obtained so far shall be described as a preliminary report.

a) Formant bandwidths of vowels

Some 70 vowel samples recorded for two male subjects have been matched by the analysis-by-synthesis scheme described in a previous report⁽⁴⁾. The data for the first and the second formants are compiled in Fig. I-A-1.

In Fig. I-A-1(a) each dot represents a recording of a vowel articulation. The three-dB-down formant bandwidth (ordinate) is related to the formant frequency (abscissa). The articulations were intended for various front, middle and back vowels with different mandible positions and also labial articulations.

The relation between the bandwidth and frequency appears to be comparatively simple for the first formant. There is a marked tendency that the bandwidth increases as the frequency decreases (towards the left of the figure). In other words, close vowels show much wider bandwidths, typically 70 c/s for [i], [y] and [u]*, than semi-open vowels, typically 35 c/s for a neutral vowel [3].

This high dissipation of the low first formant is explained by considering an appreciable participation of the surrounding soft tissues in the vibration of the acoustical system. In the low frequency region, the wall acts as a highly dissipative mass connected in parallel to the mouth outlet + radiation load⁽²⁾. The contribution of this wall vibration to the bandwidth of the formant is greater when the impedance looking toward the outside through the mouth opening is higher, which is the case for closer vowels. In these cases, other sources of dissipation, the heat conduction loss and the frictional loss of the wall are estimated to be small compared to the above-mentioned, and the radiation loss at the orifice

*Our previous report⁽⁴⁾ must be amended with a recent addition of data for very close vowels.

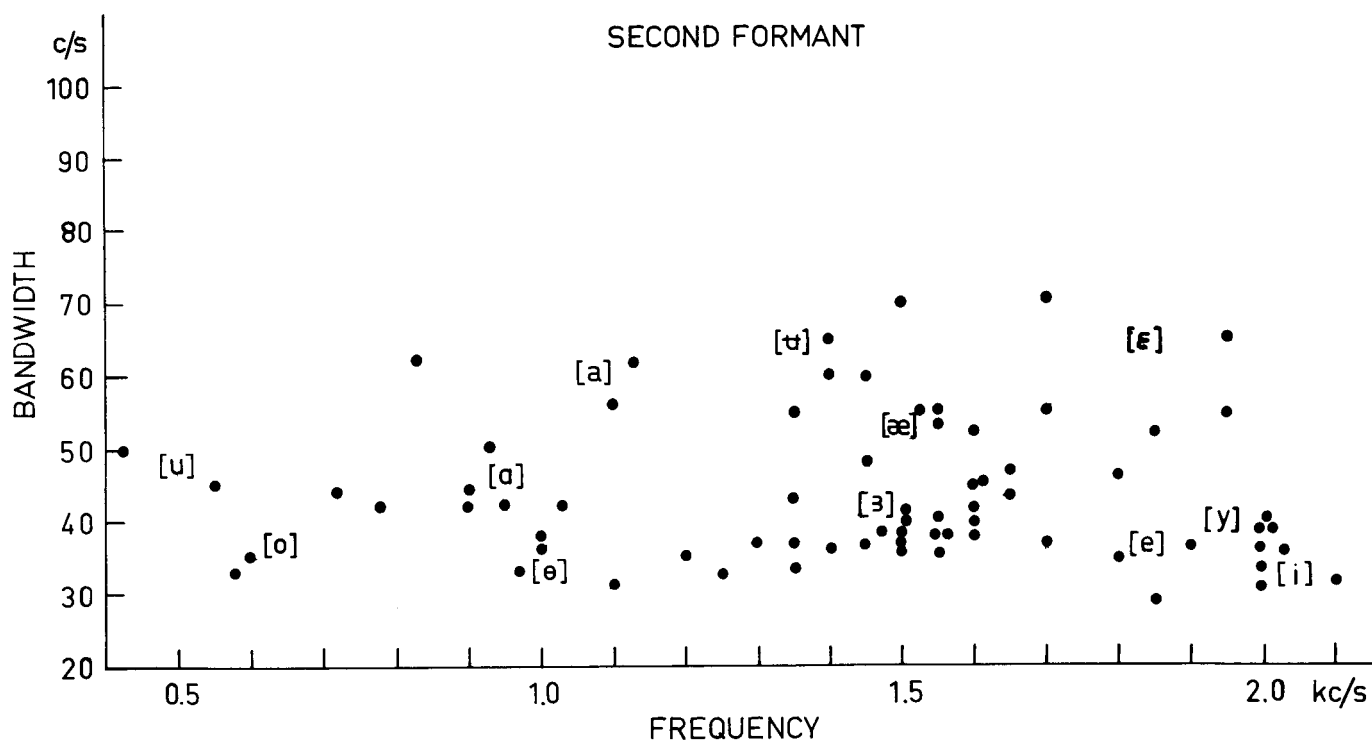
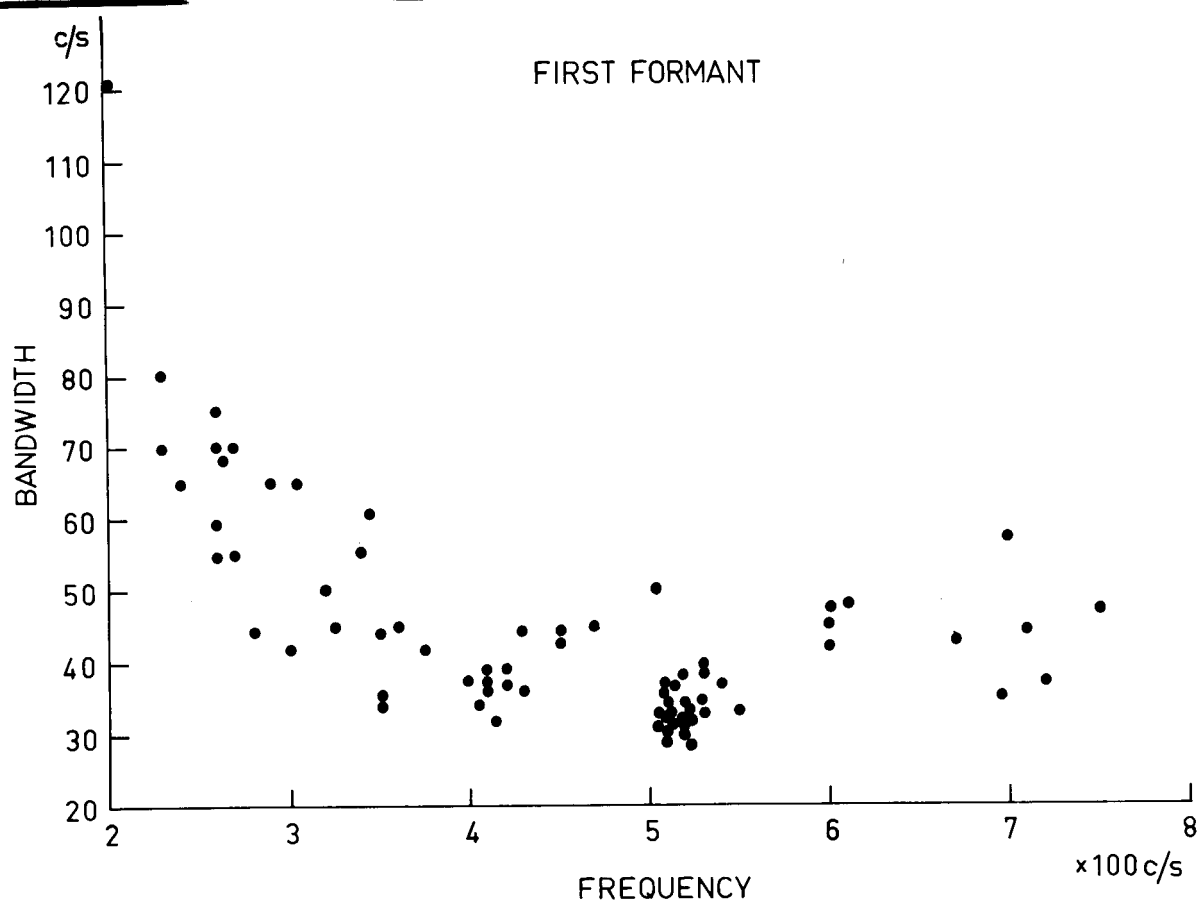


Fig. I-A.1. Bandwidth values obtained for various vowel articulations (by two male subjects).

- (a) For the first formant;
- (b) for the second formant.

is negligible. When the first formant is high in frequency, the radiation loss is expected to contribute several cycles per second to the bandwidth (see Table 2.34-3 of Reference⁽¹⁾). This would probably explain a slight rise of the bandwidth toward the right of the figure.

In Fig. I-A-1(b) similar plottings are given for the second formant. In this case, the bandwidth apparently depends more heavily on the particular vocal tract configurations, even for similar formant frequencies. It may be mentioned that close front vowels have very sharp resonances for this formant, a bandwidth value of 30 to 40 c/s for a frequency near 2000 c/s. Phonetic symbols in the figure identify representative vowel articulations for the plots in the regions.

It is perhaps worth-while to remark on the dot plotted at the left extreme of the figure. This pertains to an articulation of a strongly rounded [u] by a tall Swedish subject. The second formant is located at a frequency as low as 427 c/s. The first formant frequency for this articulation was 290 c/s. These values were accurately estimated by an analysis-by-synthesis. The highest value of the second formant frequency observed for the same subject was 2100 c/s for an articulation of [i]. The range of variation for this formant frequency is thus well above two octaves. To handle Swedish properly seems to be a nice problem for the formant tracking engineer.

Another interesting case of the frequency response is shown in Fig. I-A-2. This was recorded by the same Swedish subject for a front rounded vowel [y]. The sound was articulated with an extreme tense front tongue articulation and a strong lip-rounding. This does not represent a typical Swedish vowel, but merely one possible vowel articulation which lies well within the category of the IPA symbol.

The second and the third formants of [y] Fig. I-A-2 are located very close to each other, and this proximity of the formants results in an apparent single peak even in this continuous response curve. The analysis-by-synthesis matching naturally reveals that this peak consists of two formants, but an exact estimation of the frequencies and bandwidths is rather difficult, partly because of the limitation in the time

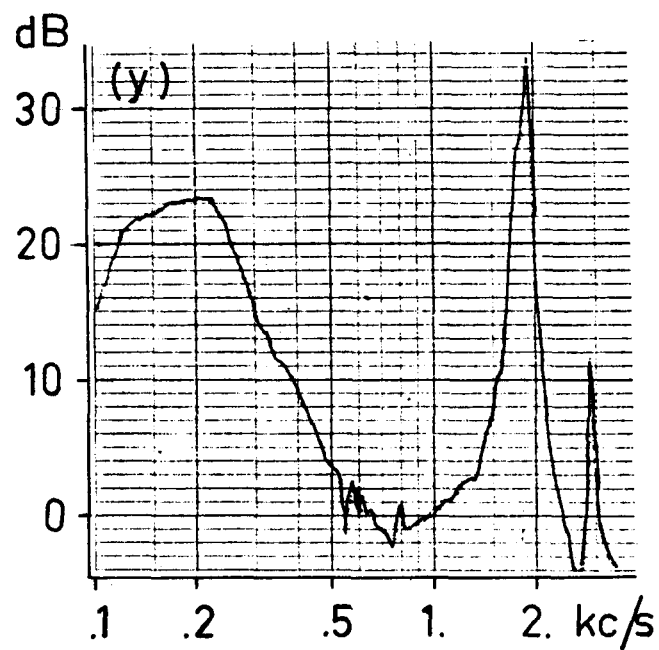


Fig. I-A-2. An example of the recorded response curve for a front rounded vowel. Note the merging of the second and the third formants into one apparent peak.

response of our recording system. It is highly probable, however, that at least one of the two formants is comparatively heavily damped.

b) Open glottis conditions

The data above pertain to a special condition of the closed glottis. Bandwidth data for natural speech data would be expected to be somewhat larger because of the partially open glottis during the phonation. It is hard to estimate the dependence of the bandwidth on various degrees of opening of the glottis. Wide open conditions for the glottis have been tried by one of our subjects, and the curves are given in Fig. I-A-3. The curve (a) is for the regular condition of the glottis and the source location. In the curve (b), the glottis was kept wide open. A significantly higher damping of the first formant is observed. The coupling of the sub-glottal system apparently causes some complexity of the curve, particularly in the ranges off the formant peaks. The curves (c), (d), and (e) were obtained also for the open glottis condition. In these cases, however, the vibrator was placed in a lower position. The effective location of the source was just below the glottis, and this was ascertained by observing that the sound output was markedly reduced by closing the glottis. (A converse effect is observed for the regular supra-glottal position of the source.) In order to see the reproducibility of the glottal condition, which was felt not obvious for the subject, the same effort was repeated.

The sub-glottal system is coupled through the orifice of the glottis in both cases, i. e. (b) and (c), (d) or (e). The poles consequently should be the same. The zeros are different, and a very effective zero can be observed near 1000 c/s in the case of the sub-glottal excitation. Evidently, this is caused by a tuning of the sub-glottal system.

c) Stops

As an extreme case of the narrow vowels, we can apply our method to a study of stop consonants, where we have a stationary and complete closure of the vocal tract at the point of articulation. In the case of natural utterances, it

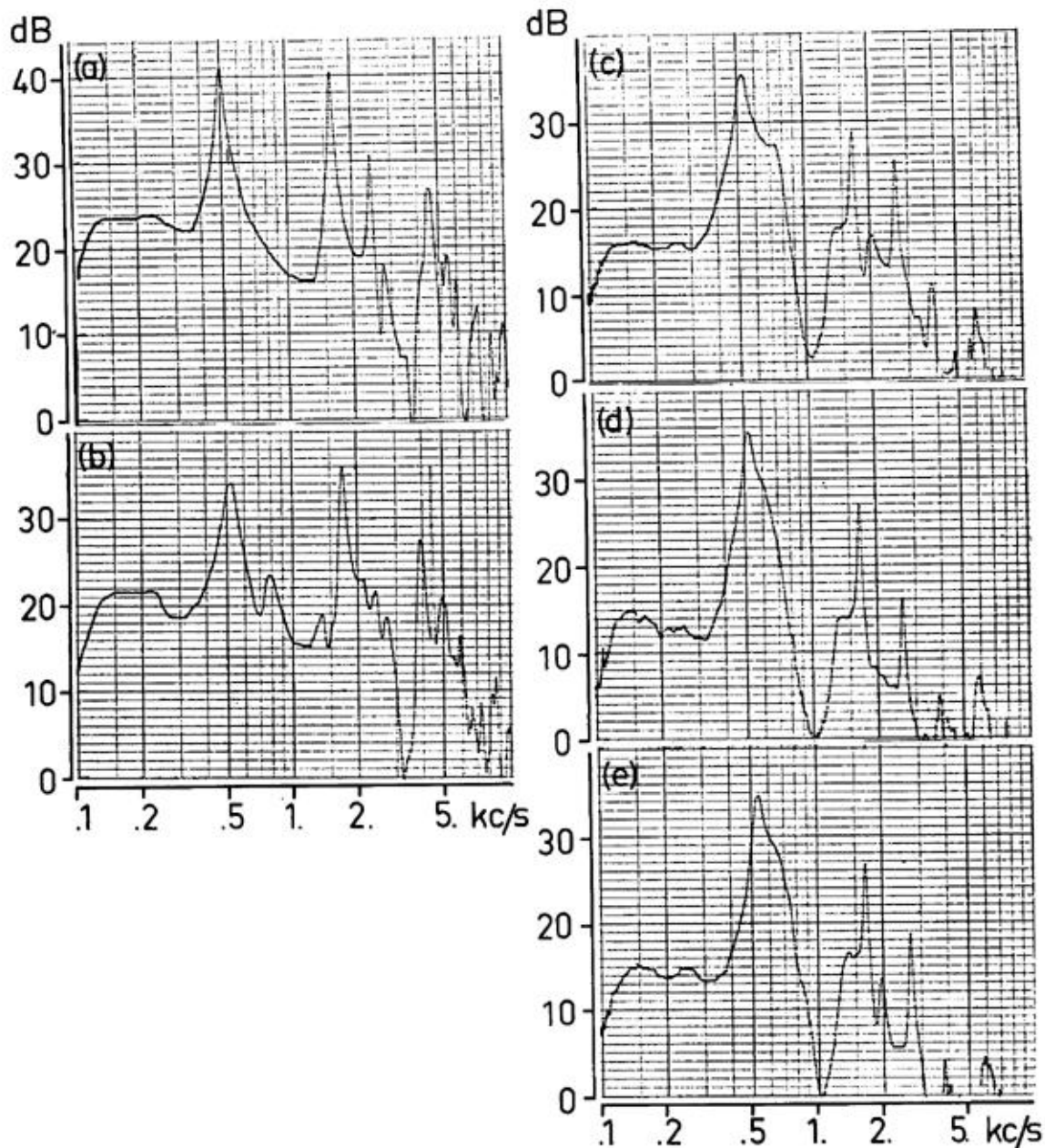


Fig. I-A.3. Frequency response curves for a neutral vowel recorded by the sweep-tone technique.

- (a) Regular response curve with the closed glottis (supra-glottal source);
- (b) open glottis, supra-glottal source;
- (c) open glottis, sub-glottal source;
- (d) same as (c), repeated;
- (e) same as (d), repeated.

is known that the spectrogram of the outcoming sound has a pronounced low frequency band, which is often called a buzz bar. The structure in higher frequency regions is usually not well observed in spectrograms. It should be kept in mind, however, that we can not assume the source characteristic to be the same as that for vowels, because in this peculiar situation of voicing, the loading condition for the vocal vibration is extremely different. With our external source, there is no such problem. The present study has revealed that the level difference between the first and the second formants is not necessarily greater in comparison to that for vowels, because the high formant can be quite sharp.

Fig. I-A-4(a) shows two samples of recorded curves of the bilabial stop. These data were recorded in the same way as described for vowels⁽⁴⁾, except that in this case the front grid of the condenser microphone was placed in contact with the skin of the cheek of the subject, in order to keep the distance between the skin and the vibrating membrane very small and constant. The second formant as well as the first can be clearly observed. The first formant has a frequency of 200 c/s or slightly lower, depending on the tongue position, for this consonant. (A slight boost to the left of this formant is due to the lowest resonance of our vibrator, see Fig. I-A-5(e).) The bandwidth has a value near 100 c/s. Many similar curves for different pick-up conditions have been tried, and it can be concluded that a first formant frequency of 170-200 c/s and a bandwidth of around 100 c/s apply to [b]'s. This conforms quite well with Fant's theory of high-pressure vowels⁽²⁾.

In the case of [d], the first formant was most often observed around 230 c/s, but a frequency of as high as 300 c/s was not unusual for the same (male) subject. In the case of a front palatal stop, the frequency has been so far estimated at 300 c/s. The velar stop is hard to measure, because of the lack of a clear formant peak. The relative peak levels for the first formant of these stops, after correction of the constant characteristics (i. e. (e) minus (f) of Fig. I-A-5), are (in reference to average [b]'s) : dental, +2 dB; palatal, 0 dB; velar, -6 dB. Dentals also show a narrower bandwidth

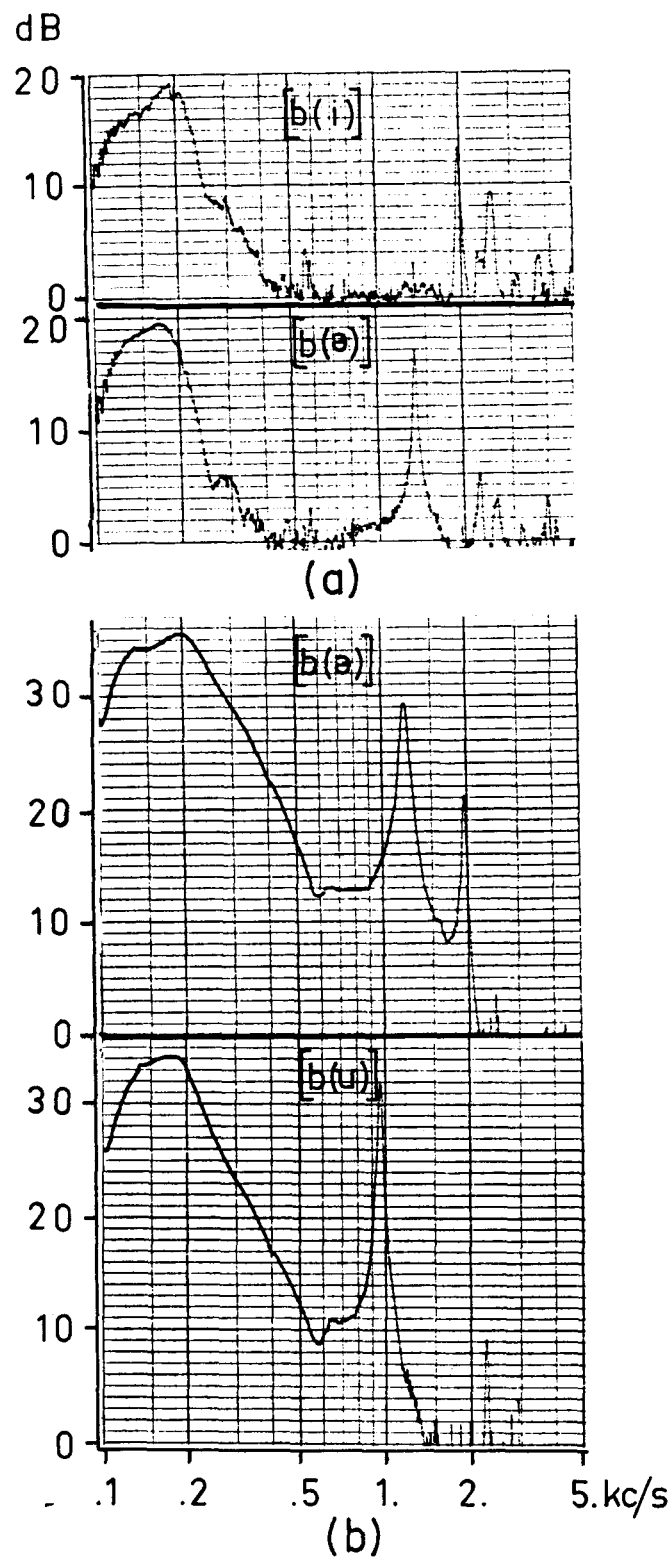


Fig. I-A-4. Response curves for the bilabial stop with different tongue coarticulations.

- (a) Recorded by a regular condenser microphone placed in contact with the cheek;
- (b) recorded by a probe microphone inserted within the labial closure.

compared to labials and palatals (typically 70 c/s).

For an estimation of formants of the bilabial stop consonant, it is possible to obtain clearer curves by inserting the tip of a thin probe microphone inside the lip closure. Fig. I-A-4(b) shows examples of the curves recorded in this way.

d) Nasalization of vowels

The effect of nasalizing a vowel has been discussed by some investigators⁽¹⁾⁽⁶⁾⁽⁷⁾. In contrast to the complexity we encounter in the case of nasal murmurs⁽⁴⁾, the response curves of nasalized vowels can be matched even in details in the lower frequency region when we assume a theoretical model of a formant-antiformant-formant structure⁽³⁾.

In one recording session, a subject articulated non-nasalized and nasalized versions of a neutral vowel [3] alternatively. The session contained seven samples of the oral vowel and six sample of the nasalized vowel. Preliminary matchings of the oral vowel curves, by use of the regular passive POVO, derived a constant correction function⁽⁴⁾, and a program disc⁽⁵⁾ of this curve was made. When this was used in rematching the vowel samples, matches were good and generally within ± 1 dB for most of the samples over the range from 100 to 2500 c/s. Some samples had local deviations of up to 2 dB. Samples located closely within the session showed better agreements among themselves.

A typical pair of consecutive samples are shown in Fig. I-A-5(a) and (b). The curve (a) represents a regular oral articulation. By an attempt of nasalizing the same vowel without changing the tongue articulation, the curve (b) (solid line) was obtained. The two curves, as well as other samples in the same session, were matched with the same correction characteristic, and satisfactory matches were obtained with the following locations of the poles and zero (in c/s, bandwidth in paronthesis):

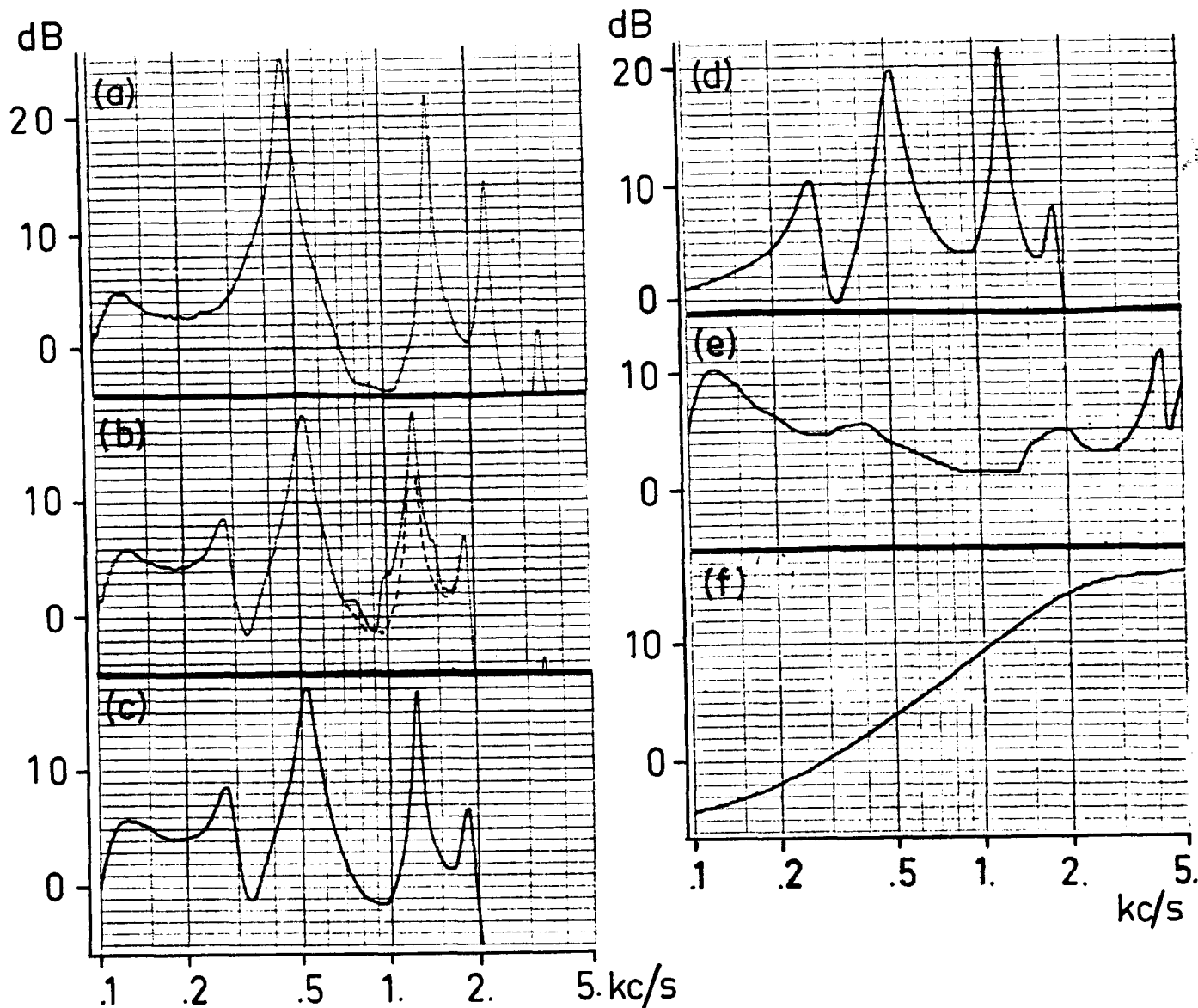


Fig. I-A-5. Data of matching articulations of a neutral vowel with and without nasalization.

- (a) Recorded response curve for [3];
- (b) same for [3̃] (the broken line is an overlaid tracing of (c);
- (c) synthesized curve for [3̃];
- (d) same without the correction curve, viz. (c) - (e);
- (e) the correction curve used in matching both the non-nasalized and nasalized vowel articulations;
- (f) high emphasis characteristic for the recording of data.

	pole 1	pole 2	pole 3	pole 4	pole 5	KH
[3]	480 (33)	1450 (42)	2300 (62)	3500	4000	4000
[3]	517 (56)	1230 (55)	1830 (154)	2700	5000	4000
	extra pole	zero				
[3]	- -	- -				
[3]	270 (46)	312 (60)				

The curve (a) was matched without any appreciable deviation, and with the same correction curve, the curve (b) was matched perfectly well except in the frequency range from 700 to 1700 c/s. The mismatch is indicated by the overlaid broken line which is identical with the curve (c). The curve (c) shows the synthesized curve, with the settings of the parameters given above. Our correction curve for this session is shown in Fig. I-A-5(e).

When we subtract this curve (e) from (c) we obtain (d). This can be considered as the transfer function of the articulatory system with a simplification in the second formant region. (This curve as well as (c) does not contain the usual +6 dB/oct characteristic of the radiation. Also, the dB-scales in this figure do not refer to any absolute level, except that they give the correct level relation between the curves (a) and (b).) The curve (f) is the high frequency pre-emphasis employed in the recording session (not used in the synthesis). Thus a subtraction of (f) from (e) would lead to an evaluation of the characteristics of our vibrator and the transmission through the wall, plus an effective transfer function relating the effective output volume velocity to the sound pressure at the microphone, which was located at front of the mouth, about 2 cm apart from it.

A perfectly good match as seen in the curve (b) assures us that the formant-antiformant-formant (F_n - A_n - F_{n+1} , see Reference⁽³⁾) structure is satisfactory for describing the nasalized vowel in the lower frequency region. The match in the second formant region seems not bad, when the damping factor is appropriately chosen, but this may not be quite true for other tongue articulations. For nasalized front vowels, in particular, it is usual to see a rather complex shape of the next higher peak above the first formant. Comparisons of oral

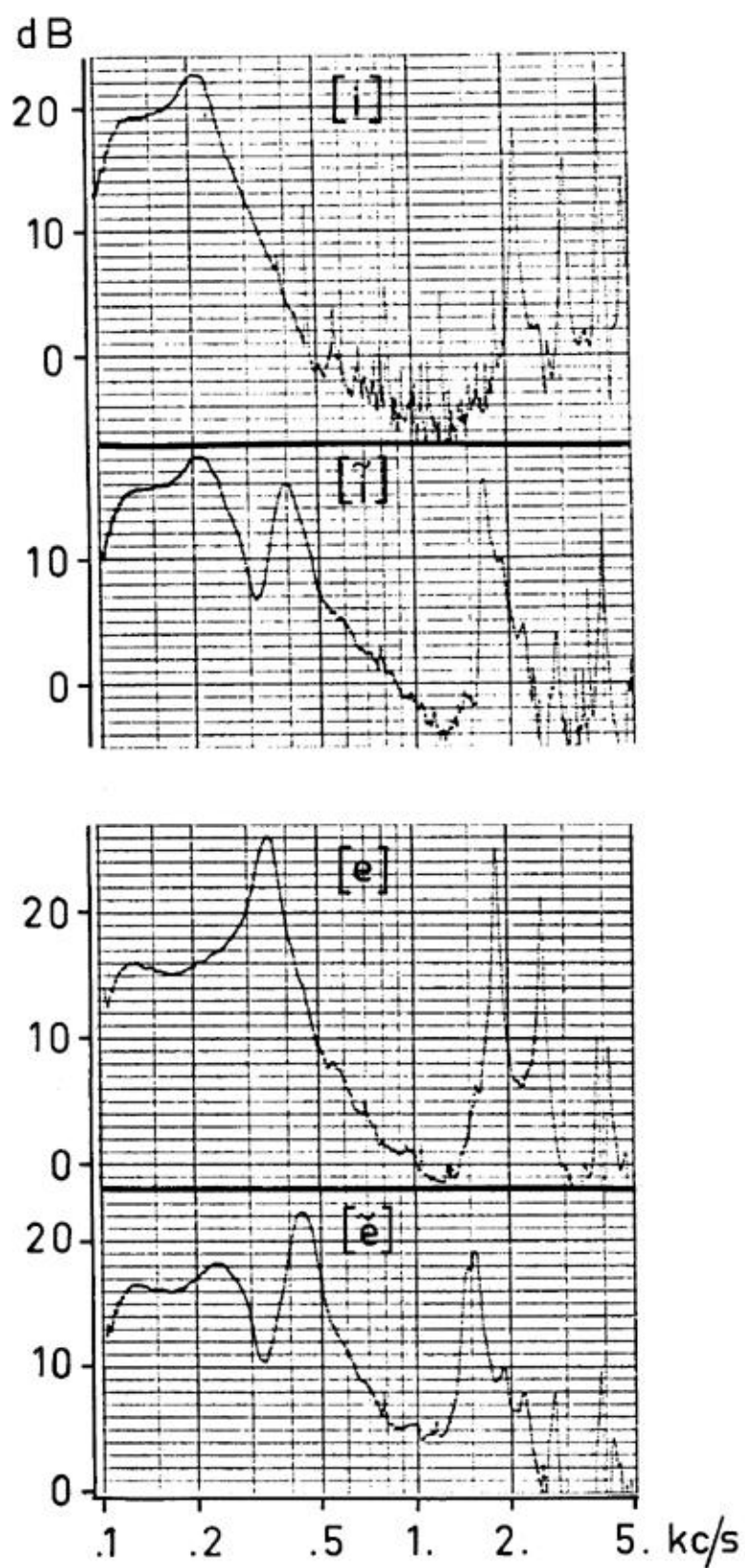


Fig. I-A-6. Comparisons of response curves for non-nasalized and nasalized articulations of front vowels.

and nasalized front vowels are exemplified in Fig. I-A-6.

In these cases, it is not obvious that the higher peak can be represented effectively (from a perceptual point of view, say) by a single formant. To the contrary, the peak rather seems to reveal a complex structure of poles and zeros, and this is not surprising, in this high frequency range, when the complex structure of nasal murmurs is recalled⁽⁴⁾.

O. Fujimura and J. Lindqvist

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